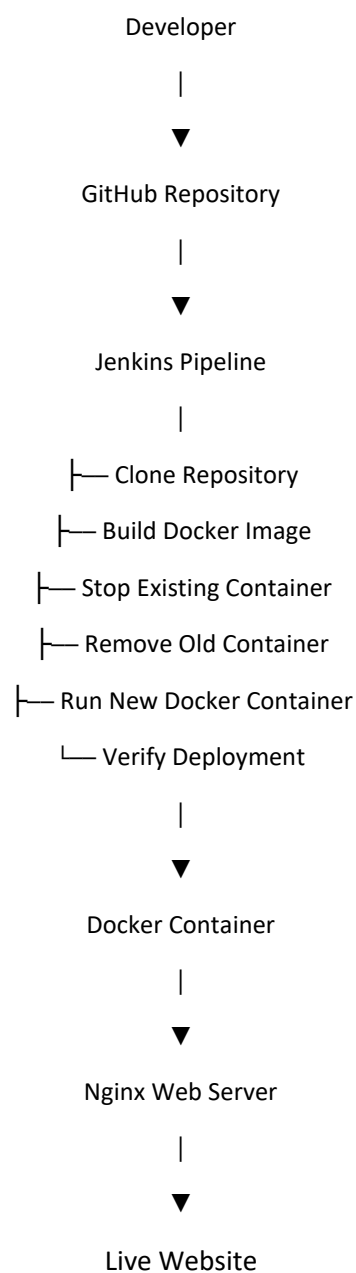


## CI/CD Pipeline using GitHub + Jenkins + Docker + Nginx

Designed and implemented an end-to-end CI/CD pipeline by creating a dedicated project repository in GitHub, configuring Jenkins for automated builds, using Docker for application containerization, and deploying the application through an Nginx web server.

### Process Flow of Project



## **STEP 1: Creation of GITHUB Repository**

## **STEP 2: Create an EC2 instance**

- Select Ubuntu OS or Amazon Linux
- In Security Group take SSH , HTTP , HTTPS, 8080 port & Key pair

## **STEP 3 : Install & Package install for Jenkins Set-Up**

- **Java JDK** installation
- **Jenkins** installations – Proceed with the below link

<https://www.jenkins.io/doc/book/installing/linux/>

After installing Jenkins check its active status , enable and start the service with below commands

```
systemctl enable jenkins
```

```
systemctl start jenkins
```

```
systemctl status jenkins
```

## **STEP 4: Docker Installation**

```
sudo apt install docker.io -y
```

```
sudo systemctl restart docker
```

```
sudo systemctl restart Jenkins
```

```
docker --version
```

## **STEP 5: Will create Jenkins user in Docker**

```
sudo usermod -aG docker
```

```
jenkins sudo usermod -aG docker ubuntu4
```

## **STEP 6: Git Installation**

```
sudo apt install git -y
```

```
git --version
```

### **Create a Folder where in to Keep the project files**

```
mkdir website-cicd
```

```
cd website-cicd
```

```
vi index.html [create file inside the dir]
```

```
vi dockerfile
```

IN DOCKER FILE INPUT BELOW COMMANDS:

**FROM nginx:latest**

**COPY . /usr/share/nginx/html**

**EXPOSE 80**

### **Create Jenkins Empty File**

```
vi jenkinsfile (empty file)
```

### **Now initialize GIT and add the contents , commit it and push the files created in GITHUB Repository**

```
git init
```

```
git add .
```

```
git commit -m "Initial Commit"
```

```
git branch -M main
```

```
git remote add origin https://github.com/yourusername/Website-CICD.git
```

```
git push -u origin main
```

username: github username

password: git token **(Create a personal token classic for password access in terminal)**

## STEP 4. Configure Jenkins

public ip of ec2

access public ip:8080 on browser

cat -var path [ copy password and then enter on Jenkins web page ]

- put credentials > save and confirm

Start using Jenkins -> Manage Jenkins -> Plugins

Install Git , Docker, Jenkins , Docker Pipeline , Pipeline , Blue Ocean (Optional)

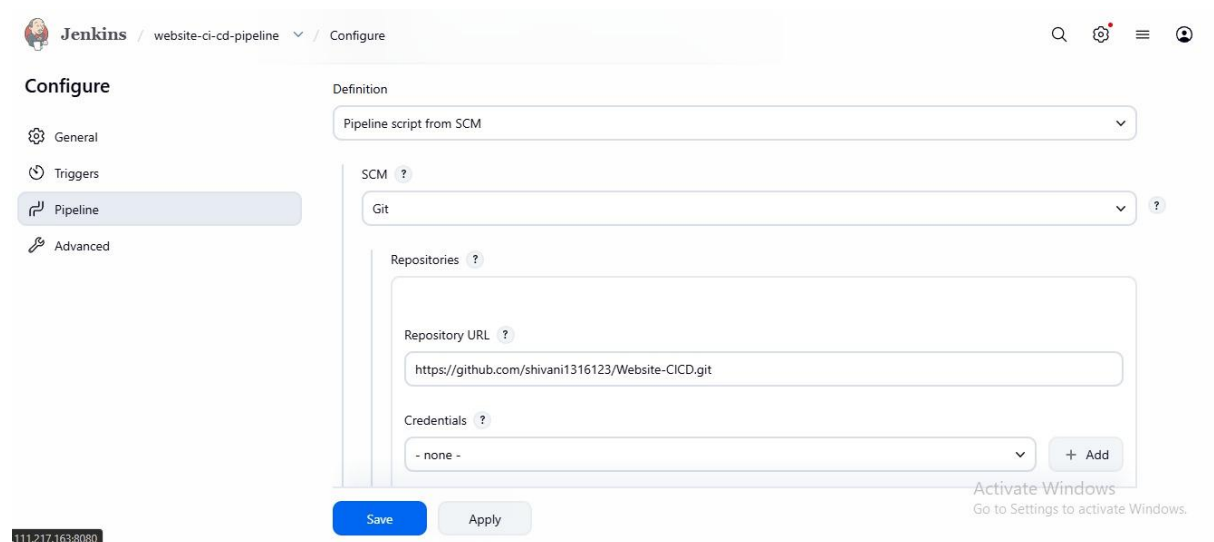
Then Restart Jenkins -> Sign in

New item > enter name: web ci-cd -> pipeline ->

General -> Pipeline -> def [ pipeline script from scm] -> SCM : Git -> Repo:  
enter git repo -> branch

Specifier : <\*/main >

Script path : Jenkinsfile



The screenshot shows the Jenkins web interface for configuring a new pipeline. The breadcrumb navigation at the top reads "Jenkins / website-ci-cd-pipeline / Configure". On the left, the "Configure" section is expanded, showing options for "General", "Triggers", "Pipeline" (which is selected), and "Advanced". The main configuration area is titled "Definition" and contains a dropdown menu set to "Pipeline script from SCM". Below this, the "SCM" dropdown is set to "Git". The "Repositories" section contains a "Repository URL" field with the value "https://github.com/shivani1316123/Website-CICD.git" and a "Credentials" dropdown set to "- none -". At the bottom of the configuration area are "Save" and "Apply" buttons. A watermark for "Activate Windows" is visible in the bottom right corner.

**\*\*\*\* If we didn't create Jenkins file before pushing the files over git follow below steps to push the file over the git hub**

-inside the dir

-vi jenkinsfile

-git add Jenkinsfile

-git commit -m "Added Jenkins pipeline"

-git push origin main

-restart Jenkins

**\*\*\*IMP**

**In Jenkinsfile make some changes if pipeline failed.**

After changes again Add , Commit and Restart Jenkins

## **STEP 5. ACCESS PUBLIC IP**

<http://Public IP>:80

